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Establishment of a native bunch grass and an invasive perennial on disturbed land using straw-amended soil

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ABSTRACT

Native grasslands around the word face increased threats from non-native species. Fescue prairie in North America, in good rangeland condition, is dominated by the perennial bunch grass, *Festuca hallii*, whereas disturbances are often colonized by *Poa pratensis*, an introduced perennial rhizomatous grass which is competitive in nitrogen rich soils. *F. hallii* thrives in typical low nitrogen grassland soils and recovers poorly once disturbed. Disturbance to soil caused by well site construction may decrease organic carbon and potassium, and increase nitrogen, phosphorus, pH and electrical conductivity, creating conditions conducive to invasion by *P. pratensis*. This research tested the hypothesis that *F. hallii* would tolerate nitrogen depleted soil, through addition of carbon as a straw amendment to newly reclaimed well sites, better than *P. pratensis*. Our second hypothesis was that *F. hallii* is negatively affected by disturbed soil and *P. pratensis* is not. We treated three sites with three straw amendment rates, seeded monocultures of *F. hallii* and *P. pratensis*, and monitored establishment over three years. *F. hallii* biomass, root biomass, leaf length and cover increased in response to straw treatments, whereas *P. pratensis* showed little response. *F. hallii* was positively affected by prior-year soil water, and current-year ammonium and potassium. *P. pratensis* was positively affected by current-year soil water, potassium and nitrate. *P. pratensis* responded positively to higher pH and electrical conductivity found in disturbed soil and *F. hallii* responded poorly. The positive relationship of *P. pratensis* to pH above 7 could explain why it can invade reclaimed disturbed grassland; whereas the negative reaction of *F. hallii* might explain its failure to recover. We concluded the addition of straw as a soil amendment is a possible solution to poor establishment of *F. hallii*.

1. Introduction

Native grasslands world-wide are increasingly threatened by invasion of non-native species (Romo et al., 1990; Samson and Knopf, 1996; Forrest et al., 2004). Grassland portions of the aspen parkland in western North America were originally fescue prairie, dominated by *Festuca hallii* (Vasey) Piper (rough fescue) (Coupland, 1961). Oil and gas development, commencing in the 1940s, combined with cultivation, removed all but small remnant parcels of fescue prairie throughout the area (Grilz et al., 1994). *F. hallii* recovers poorly, once disturbed, and fescue prairie is often colonized by *Poa pratensis* L. (Kentucky bluegrass) to the exclusion of native species (Brown, 1997; Slogan, 1997; Desserud, 2006).

F. hallii, a perennial bunch grass, is slow growing and long lived, requiring two to three years to become established from seed. It seldom produces seed, with several or many years between seeding events (Johnston and MacDonald, 1967; Toynbee, 1987; Romo, 1996). *P. pratensis* is an exotic grass in North America, presumably arriving with European settlement (Tannas, 2001). It is now naturalized in Alberta and often establishes from the seed bank when soil is disturbed (Adams et al., 2005). Unlike *F. hallii*, *P. pratensis* readily produces seed in its first and subsequent years, and is strongly rhizomatous, allowing it to establish and spread rapidly (Best et al., 1971; Tannas, 2001).

Disturbance to soil caused by oil and gas well site construction may initially increase nitrogen availability (Dormaar and Willms, 1990) due to organic matter mineralization. Mixing subsoil with topsoil may decrease soil organic carbon (C) and potassium (K), and increase clay content, phosphorus (P), pH and electrical conductivity (EC) (de Jong and Button, 1973; Culley et al., 1982; Naeth et al., 1987; Hammermeister et al., 2003). These effects alter the

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competitive relationships among plant species and may drive succession in an undesirable direction.

In grasslands, increased availability of nitrogen is known to promote invasion of non-native plants, as demonstrated in Great Britain (Burke and Grime, 1996), and in the United States in Minnesota (Wedin and Tilman, 1996), Colorado (Milchunas and Lauenroth, 1995) and Hawaii (Vitousek and Lawrence, 1989). If soil nitrogen facilitates non-native plants, then reducing soil nitrogen, such as through carbon addition (McLendon and Redente, 1992; Morgan, 1994; Davis and Wilson, 1997), may inhibit their establishment. Non-native plants were reduced by addition of carbon to disturbed grasslands with sawdust in northern California (Alpert and Maron, 2000; Alpert, 2010), sucrose in western Utah (Beckstead and Augspurger, 2004) and straw in the Netherlands (Kardol et al., 2008).

One method to aid recovery of fescue and other native grasslands, and reduce invasion of non-native plants, may be addition of straw as a carbon soil amendment. Straw may alter soil properties through its decomposition by adding C, K (Christensen, 1985) and ammonium (NH_4^+) (Smith and Douglas, 1971). Straw may also provide mulch, facilitating soil water retention (Johnston, 1961a). Whereas carbon amendments may negatively affect non-native species, there appears to be little effect on native grasses (Alpert and Maron, 2000).

Finding a competitive advantage for *F. hallii* could lead to an effective reclamation technique to assist its recovery following disturbance. This research tested the hypothesis that *F. hallii* will benefit from addition of straw and resulting soil chemical changes. Nitrogen depletion through addition of carbon in the form of straw may reduce establishment of *P. pratensis*. The second hypothesis is that *F. hallii* is negatively affected by disturbed soil, which results in lower organic C, K, P, pH and EC, whereas *P. pratensis* may tolerate such soil changes. Past failures of *F. hallii* recovery and successful invasion of *P. pratensis* after disturbance could be explained by understanding differences in soil and vegetation properties.

2. Study area

Three field sites were established in 2007 in central Alberta, Canada: two in the Central Parkland natural region at Ellerslie (53° 25′ N, 113° 29′ W) and Byemoor (51° 59′ N, 112° 19′ W) and one in the Northern Fescue subregion of the Grassland natural region at Drumheller (51° 26′ N, 112° 21′ W). Elevation is approximately 692, 843 and 929 m above sea level, respectively. Soils at Ellerslie are Orthic Black Chernozems and at Byemoor and Drumheller they are Dark Brown Chernozems. Native grassland vegetation in both regions is dominated by *F. hallii* associated with *Hesperostipa curtiseta* (A.S. Hitchc.) Barkworth (western porcupine grass) in the Central Parkland, and *Bouteloua gracilis* (Willd. ex Kunth) Lag. ex Griffiths (blue grama grass) in the Northern Fescue subregion.

During the research period between June 2007 and July 2009, temperature maximums were 36 °C, with growing season (April to October) temperatures averaging 12 °C. Average annual rainfall in this period was 510 mm at Ellerslie, 302 mm at Byemoor and 270 mm at Drumheller. The Ellerslie site was 1 ha in size and located in a previously cultivated area. Prior to our experiment it was fallowed for 3 years, and in May 2007 it was sprayed with glyphosate (Roundup Weathermax®) at 0.4 1/ha with a Flexicoil sprayer. Topsoil was approximately 23 cm deep. The Drumheller site was an abandoned natural gas well site (0.5 ha) in uncultivated land. In June 2007 it was cleared of topsoil, the topsoil stockpiled and stored on-site for less than a week. Flow-line piping was removed, the well capped and topsoil replaced to a depth of approximately 11 cm. The Byemoor site (1 ha) was in uncultivated land. The site was leveled and cleared of topsoil in preparation for gas well drilling in 2006, with the topsoil stockpiled and stored on-site for one year. It was recontoured in 2007 and topsoil replaced to a depth of approximately 30 cm.

3. Methods

3.1. Experimental design

In July 2007, the three sites were treated with straw amendments and seeded. Seeding and straw treatments were blocked as 96 factorial random experimental subplots at Byemoor and Ellerslie: 3 (straw treatments) × 4 (reps) × 2 (species) × 4 (reps) and 48 at Drumheller: 3 (straw treatments) × 4 (reps) × 2 (species) × 2 (reps) (Fig. 1). Seeded vegetation was sampled, as detailed below, in late July and early August 2008 in randomly selected subplots (6–7 m^2); the same subplots were sampled in 2009.

During sampling, several subplots at each site were discarded due to species contamination or erosion: subplots at the southeast edge of Byemoor were affected by riparian species originating from a nearby wetland; the complete western subplot row of Ellerslie was invaded by agronomic species from a nearby crop field; and plots in the southwest corner of Drumheller were eroded by wind and water, exposing an ancient river bed heavily covered with stones. This resulted in lowering the number of potential sampling subplots: 84 at Byemoor and Ellerslie. Thus samples were randomly selected in the un-compromised subplots of each seeding by straw treatment: 42 subplots at Byemoor and Ellerslie and 30 at and at Drumheller (Fig. 1). All sampling incorporated a 50 cm wide buffer around each subplot to avoid edge effects.

3.2. Straw treatment

One-year-old straw was applied at three rates: approximately 1 kg/m^2 (high), 0.5 kg/m^2 (low) and no straw (control). Large round bales of straw (700–800 kg) were chopped with a bale processor running at approximately 2 km/h. Wheat straw was applied at Ellerslie and Byemoor, and barley straw was applied at Drumheller. Chopped straw was spread in strips 6–7.5 m wide and 72–90 m long (Fig. 1). Straw was incorporated 8–10 cm into the soil with

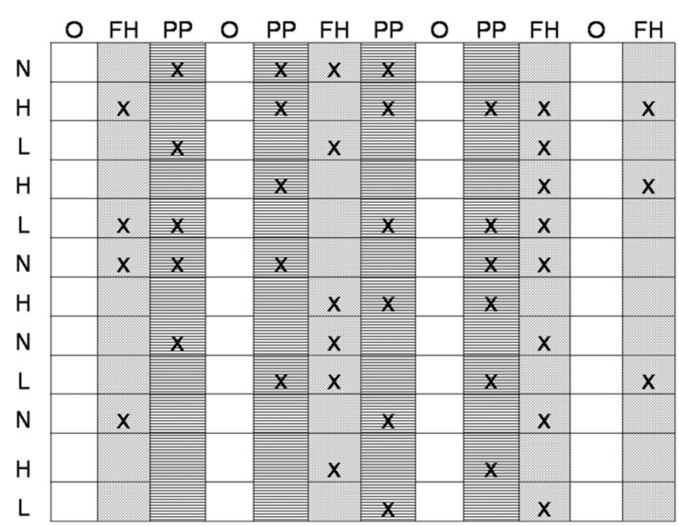

Fig. 1. Schematic of experimental design showing factorial random subplots of seeding and straw treatments. FH = *Festuca hallii*, PP = *Poa pratensis* seeding, H = high straw (1 kg/m^2), L = low straw (0.5 kg/m^2) and N = no amendment. O = other seeding, part of another study. At Byemoor and Ellerslie each row was 7 m wide, and at Drumheller each row was 5 m wide. At each site, 5–7 subplots of each seeding/straw combination were randomly selected for vegetation and soil sampling.

a 2 m-wide rototiller pulled by a 50 hp tractor running at 1800 rpm and 2.4 km/h. Control strips were also rototilled. Four replications were laid out in a fully randomized design at Ellerslie and Byemoor (Fig. 1). However, to retain equivalent plot sizes only two replications of straw were laid out at Drumheller.

3.3. Seeding treatment

Seeding occurred in July 2007 within one week of straw application. Strips of *F. hallii* (15.5 kg/ha, 1300 live seeds/m) and *P. pratensis* (1.7 kg/ha, 775 live seeds/m) were seeded perpendicular to and across the straw amendment treatments (Fig. 1). A third species treatment was applied to one-third of the site, not applicable to this research and described elsewhere (Desserud, 2011). *F. hallii* seed was collected in July 2006, in rough fescue grassland in the Rumsey Natural Area (51° 50′ N, 112° 33′ W), in central Alberta with a combine set at a height of approximately 30 cm. Seed was air dried at 25 °C, the chaff removed and *F. hallii* seed separated from other grasses. To test seed viability, 100 randomly selected seeds were distributed equally among ten closed petri dishes on 1 mm thick germination paper, wet with distilled water and incubated in the dark at 20 °C (Romo et al., 1991). Ninety-five percent of the seeds germinated (initial cotyledon shoot length 0.5–1 cm) within 6 weeks; therefore, the collected seed was deemed viable. *P. pratensis* seed was obtained from a commercial seed supplier with certified seed viability. Seed was placed at a depth of 1.2–1.9 cm with a seeder and packer with double disk openers.

3.4. Vegetation analyses

Within each sampled subplot, up to five specimens of seeded *F. hallii* or *P. pratensis* were randomly selected and measured for leaf length, number of inflorescences and vegetative tillers. In each sampled subplot, three randomly selected specimens of the seeded species were excavated with a hand shovel. Plants with intact roots were stored in plastic ziploc bags at 5 °C in a refrigerator until washing. Whole plants were washed with running tap water, roots and crowns were removed and washed thoroughly to remove all traces of soil. Leaves and roots with crowns were oven dried, separately, oven dried for 48 h at 60 °C then weighed to determine dry biomass.

3.5. Soil analyses

Soil was sampled adjacent to the excavated plants, three locations per subplot, in 2008 and 2009, to a depth of 10 cm using a hand auger (5 cm diam.). The three samples of each subplot were composited, resulting in 42 soil samples from Byemoor and Ellerslie and 30 from Drumheller. Volumetric soil water was measured in June, July and August in 2008 and 2009 at three random locations per subplot, using a ThetaProbe ML2x® at 0 and 15 cm depths, resulting in 126 soil water measurements from Byemoor and Ellerslie and 90 from Drumheller.

Soil was sampled in the final year (2009) of research in undisturbed grassland within 15 m of the Byemoor and Drumheller site boundaries. Four samples were taken 25 m apart along each side of the sites with a hand auger to a depth of 10 cm. Every two adjacent samples were combined for a total of 8 samples from each site. All soil samples were stored at 5 °C in a refrigerator until they were analyzed.

Soils were analyzed for total carbon (C) by LECO combustion (Nelson and Sommers, 1996), total soil nitrogen using the Dumas method (Bremner, 1996), available ammonium (NH_4^+) using the 2.0M KCl (potassium chloride) procedure (Maynard and Kalra, 1993a) and available nitrate (NO_3^-) with dilute calcium chloride solution extraction (Maynard and Kalra, 1993b). Soil electrical conductivity (EC) and pH were measured in a 1:2 soil:water extraction (Carter and Gregorich, 1993b, 1993a). Available phosphate (PO_4^-) and potassium were analyzed by modified Kelowna extraction (Qian et al., 1994). Soil texture was determined manually (Day, 1965).

3.6. Statistical analyses

Subplots were randomly selected and data were analyzed as a fully factorial randomized design of three field sites by three treatments by two species. Data were normally distributed with Shapiro–Wilk analysis. Variations of growth responses among sites and treatments were analyzed with multivariate general linear model (GLM). Growth responses and soil variable differences among and within sites and treatments were tested by univariate analysis of variance (ANOVA) with Tukey's post-hoc test. Differences between years were analyzed with student *t*-tests. Relationships among growth responses and soil variables were assessed with Pearson product correlation and linear regression. Data analyses employed IBM® SPSS® Statistics (version 18, SPSS, Chicago, IL) and Microsoft® Excel© 2007 (Microsoft Corporation, Redmond, Washington).

4. Results

4.1. Straw responses

4.1.1. Vegetative responses

F. hallii responded positively to straw treatments; whereas, straw had little effect on *P. pratensis* (Tables 1 and 2). *F. hallii* biomass, root biomass, leaf length and cover were greater with high-straw treatments (Table 1). *P. pratensis* response to straw was opposite that of *F. hallii*, with greater leaf length and root biomass in no-straw treatments (Table 2).

4.1.2. Soil responses to straw

In *F. hallii* treatments, NH_4^+ concentrations were greatest in high-straw, decreasing in all straw treatments between 2008 and 2009, with the greatest decrease in high-straw (Table 3). Nitrate and PO_4^- concentrations were greatest in no-straw, increasing significantly ($p = 0.012$; $p = 0.015$) in all straw treatments in 2009. Potassium concentrations were initially greatest in high-straw, and had the largest decrease the following year (Table 3).

In *P. pratensis* treatments, no-straw treatments had the greatest NO_3^- concentrations ($p = 0.011$) in 2008, decreasing in 2009; whereas, PO_4^- ($p = 0.014$) and K ($p = 0.071$) increased in all straw treatments (Table 4).

4.2. Site variations

In *F. hallii* and *P. pratensis* treatments, 2008 June and total available soil water were highest at Byemoor; whereas, Drumheller had the least 2008 mid-summer soil water (Tables 5 and 6). In 2009, Ellerslie had the highest soil water levels (Tables 5 and 6). In *F. hallii* treatments, NH_4^+ concentrations dropped between 2008 and 2009 at Drumheller ($p < 0.001$) and Ellerslie, and were significantly different among sites in both years (Table 5). In *P. pratensis* treatments, NH_4^+ concentrations dropped between 2008 and 2009 and were significantly different among sites (Table 6).

Phosphate concentration was significantly higher in disturbed soils than in adjacent undisturbed soils at Byemoor ($p < 0.001$) and Drumheller ($p = 0.010$; Table 7). In *P. pratensis* treatments, Ellerslie had significantly greater C and N concentrations. In 2009 Ellerslie had the greatest NO_3^- concentrations; however, Drumheller had the

Table 1
Mean (±1 SD) vegetation properties of *Festuca hallii* and straw treatments in 2008 and 2009, analyzed by ANOVA and Tukey post-hoc tests. Letters indicate significant differences among treatments at $p \leq 0.05$.

	Year	Straw treatment			p
		High straw	Low straw	No straw	
All Sites					
Biomass (g)	2008	0.8 (0)	0.7 (0)	0.7 (1)	0.820
	2009	4.0 (2)	3.4 (2.0)	2.5 (2)	0.069
Root biomass (g)	2008	0.2 (0)	0.1 (0)	0.1 (0)	0.205
	2009	1.0a (1)	0.6ab (0)	0.6b (0)	0.040
Leaf length (cm)	2008	13.2 (5)	13.3 (5)	11.5 (4)	0.521
	2009	18.5 (5)	17.8 (5)	17.4 (6)	0.817
Cover (%)	2008	8.4 (8)	7.6 (7)	6.0 (8)	0.614
	2009	19.0a (16)	16ab (11)	9.0b (9)	0.048
Byemoor					
Biomass (g)	2008	0.7 (1)	0.5 (1)	0.4 (0)	0.444
	2009	1.7a (1)	1.2ab (1)	0.7b (0)	0.049
Root biomass (g)	2008	0.04 (0)	0.03 (0)	0.03 (0)	0.564
	2009	0.3 (0)	0.2 (0)	0.1 (0)	0.166
Leaf length (cm)	2008	7.9 (2)	8.6 (12)	8.6 (1)	0.606
	2009	12.0 (1)	12.0 (2)	11.5 (3)	0.912
Cover (%)	2008	5.7 (8)	8.9 (4)	2.2 (3)	0.302
	2009	5.8 (4)	5.3 (5)	4.1 (3)	0.754
Drumheller					
Biomass (g)	2008	0.6 (0)	0.6 (0)	1.0 (1)	0.465
	2009	5.5a (2)	3.5ab (1)	2.7b (1)	0.043
Root biomass (g)	2008	0.2 (0)	0.3 (0)	0.2 (0)	0.352
	2009	2.0a (1)	0.8b (0)	0.9ab (0)	0.050
Leaf length (cm)	2008	11.9 (4)	13.6 (3)	18.6 (5)	0.080
	2009	22.0 (5)	18.0 (3)	18.0 (4)	0.323
Cover (%)	2008	2.7 (6)	12.1 (4)	17.9 (26)	0.402
	2009	33.0a (13)	16a (8)	7.6b (4)	0.003
Ellerslie					
Biomass (g)	2008	0.3 (0)	0.2 (9)	0.2 (0)	0.264
	2009	4.9 (1)	4.9 (1)	4.0 (1)	0.411
Root biomass (g)	2008	0.3 (0)	0.2 (0)	0.2 (0)	0.264
	2009	0.9 (0)	0.6 (0)	0.7 (0)	0.136
Leaf length (cm)	2008	19.2a (2)	18.6a (3)	15.4b (2)	0.050
	2009	22.0 (1)	22.0 (2)	23.0 (3)	0.460
Cover (%)	2008	8.3 (3)	5.6 (2)	6.3 (2)	0.298
	2009	20.0 (17)	25.0 (8)	14.0 (12)	0.310

Table 2
Mean (±1 SD) vegetation properties of *Poa pratensis* and straw treatments in 2008 and 2009, analyzed by ANOVA and Tukey post-hoc tests. Letters indicate significant differences among treatments at $p \leq 0.05$.

	Year	Straw treatment			p
		High straw	Low straw	No straw	
All Sites					
Biomass (g)	2008	3.4 (4)	4.0 (4)	3.9 (6)	0.983
	2009	6.2 (3)	7.6 (5)	7.6 (4)	0.545
Root biomass (g)	2008	0.8 (1)	1.8 (3)	1.4 (3)	0.462
	2009	3.1 (2)	3.1 (5)	3.7 (3)	0.844
Leaf length (cm)	2008	29.0 (17)	40.0 (22)	40.0 (19)	0.301
	2009	32.5 (17)	35.5 (17)	33.9 (20)	0.897
Cover (%)	2008	21.0 (18)	16.0 (15)	27.2 (18)	0.176
	2009	30.0 (20)	28.0 (21)	31.0 (21)	0.859
Byemoor					
Biomass (g)	2008	1.6 (0)	2.3	1.4 (1)	0.092
	2009	7.1 (3)	9.6 (4)	11.0 (5)	0.337
Root biomass (g)	2008	0.2 (0)	0.2 (0)	0.2 (0)	0.871
	2009	2.2 (1)	5.7 (7)	5.6 (4)	0.305
Leaf length (cm)	2008	34.0 (8)	43.0 (0)	44 (14)	0.324
	2009	23.2 (6)	27.8 (7)	21.6 (6)	0.249
Cover (%)	2008	33.0 (19)	15.7 (5)	24.0 (12)	0.250
	2009	33.0 (11)	30.0 (15)	39.0 (17)	0.531
Drumheller					
Biomass (g)	2008	1.0 (1)	1.3	1.3 (0)	0.545
	2009	4.8 (2)	2.8 (2)	4.6 (3)	0.389
Root biomass (g)	2008	0.2 (0)	0.3 (0)	0.2 (0)	0.352
	2009	3.9 (3)	1.8 (2)	1.9 (1)	0.245
Leaf length (cm)	2008	11.9 (4)	13.6 (5)	18.6 (5)	0.080
	2009	22.4 (5)	21.9 (2)	21.4 (6)	0.948
Cover (%)	2008	2.7 (6)	12.1 (6)	17.9 (26)	0.402
	2009	25.5 (23)	21.3 (15)	14.8 (22)	0.692
Ellerslie					
Biomass (g)	2008	7.5 (5)	7.7 (3)	10.5 (9)	0.699
	2009	6.2 (3)	8.9 (6)	6.8 (2)	0.575
Root biomass (g)	2008	2.1 (1)	4.5 (1)	4.4 (3)	0.323
	2009	3.6 (2)	2.6 (2)	3.1 (1)	0.616
Leaf length (cm)	2008	51.2 (18)	58.8 (20)	56.2 (14)	0.850
	2009	58.0 (2)	61.0 (4)	61.0 (5)	0.455
Cover (%)	2008	28.0 (1)	19.0 (6)	39.0 (13)	0.120
	2009	32.0 (26)	30.0 (33)	39.0 (19)	0.826

greatest concentration in 2009 (Table 6). Byemoor had the highest soil pH and EC and Ellerslie had the lowest (Tables 5 and 6). Byemoor pH ($p < 0.001$) and EC ($p = 0.003$), in both seeding treatments, was higher than that in adjacent undisturbed grassland. At Drumheller, undisturbed grassland pH and EC were similar to that of the disturbed site (Tables 5–7). Of Byemoor sampling plots, 19% were clay textured, 54% were clay-loam and 27% were loam; at Ellerslie 100% were clay-loam; and at Drumheller 100% were loam.

F. hallii shoot biomass, root biomass, leaf length and cover, were lowest at Byemoor and greatest at Drumheller (Table 1). Ellerslie had the longest *P. pratensis* leaf length and Drumheller had the shortest (Table 2).

4.3. Vegetation responses to soil characteristics

F. hallii shoot biomass and cover increased with previous year July soil water (Fig. 2). *P. pratensis* leaf length increased with current-year soil water (Fig. 3). *F. hallii* 2009 shoot biomass ($r^2 = 0.14$, $p = 0.011$; $r^2 = 0.09$, $p = 0.038$) and cover ($r^2 = 0.27$, $p < 0.001$; $r^2 = 0.08$, $p = 0.049$) increased with higher concentrations of prior- and current-year NH_4^+ (Fig. 4). *F. hallii* shoot biomass and leaf length ($r^2 = 0.24$, $p < 0.001$) decreased with current-year increased K (Fig. 4).

P. pratensis shoot biomass ($r^2 = 0.10$, $p = 0.038$) and root biomass ($r^2 = 0.14$, $p = 0.015$) correlated positively to current-year NO_3 and leaf length ($r^2 = 0.24$, $p = 0.001$) correlated negatively. *P. pratensis*

Table 3
Mean (±1 SD) soil properties of *Festuca hallii* treatments in different straw treatments at all sites in 2008 and 2009, analyzed by ANOVA and Tukey post-hoc tests.

	Year	2008 straw level			p
		High	Low	None	
June soil water (%)	2008	32.2 (7)	28.6 (8)	27.3 (4)	0.265
	2009	7.4 (2)	7.5 (2)	6.3 (2)	0.153
July soil water (%)	2008	15.9 (6)	17.8 (6)	14.4 (3)	0.154
	2009	25.6 (7)	25.9 (8)	26.6 (9)	0.934
August soil water (%)	2008	9.1 (4)	8.8 (6)	7.1 (4)	0.575
	2009	19.0 (11)	19.4 (10)	20.1 (10)	0.948
C:N ratio	2008	12.0 (5)	10.6 (2)	10.2 (1)	0.271
	2009	10.4 (1)	10.4 (1)	10.3 (1)	0.630
Total carbon (C) (mg/kg)	2008	3.4 (1)	3.0 (1)	3.3 (1)	0.531
	2009	4.1 (2)	3.9 (1)	3.9 (1)	0.942
Total nitrogen (N) (mg/kg)	2008	0.3 (0)	0.3 (0)	0.3 (0)	0.918
	2009	0.4 (0)	0.4 (0)	0.4 (0)	0.962
Available ammonium (NH_4^+) (mg/kg)	2008	8.5 (13)	5.3 (4)	3.9 (1)	0.270
	2009	4.3 (2)	3.7 (2)	3.4 (2)	0.262
Available nitrate (NO_3^-) (mg/kg)	2008	6.3 (6)	5.1 (4)	8.5 (14)	0.588
	2009	8.1 (7)	7.0 (4)	11.4 (13)	0.350
Available phosphate (PO_4^-) (mg/kg)	2008	9.8 (4)	9.2 (3)	9.8 (5)	0.890
	2009	11.7 (5)	12.8 (4)	13.0 (3)	0.631
Available potassium (K) (mg/k^{10})	2008	253 (167)	210.8 (74)	210 (78)	0.301
	2009	216 (81)	219.8 (85)	219 (68)	0.988
pH	2008	7.4 (1)	7.4 (1)	7.3 (1)	0.925
	2009	7.4 (1)	7.4 (1)	7.5 (1)	0.894
Electrical conductivity (EC) (ds/m^2)	2008	0.3 (0)	0.3 (0)	0.3 (0)	0.914
	2009	0.2 (0)	0.2 (0)	0.3 (0)	0.588

Table 4
Mean (±1 SD) soil properties of *Poa pratensis* treatments in different straw treatments at all sites in 2008 and 2009, analyzed by ANOVA and Tukey post-hoc tests. Letters indicate significant differences among treatments at $p \leq 0.05$.

	Year	2008 straw level			p
		High	Low	None	
June soil water (%)	2008	31.3a (7)	31.1a (8)	27.3b (8)	0.039
	2009	5.3 (2)	6.5 (2)	6.1 (2)	0.133
July soil water (%)	2008	18.4 (6)	16.4 (4)	15.1 (5)	0.186
	2009	23.4 (8)	26.4 (8)	25.4 (9)	0.882
August soil water (%)	2008	7.5 (3)	7.6 (4)	7.9 (4)	0.345
	2009	21.8 (10)	22.7 (10)	21.4 (11)	0.942
C:N ratio	2008	10.9 (1)	11.7 (3)	10.7 (3)	0.525
	2009	10.3 (0)	10.2 (2)	10.4 (5)	0.868
Total carbon (C) (mg/kg)	2008	3.6 (1)	3.4 (1)	3.8 (2)	0.746
	2009	3.8 (1)	3.6 (1)	3.9 (1)	0.808
Total nitrogen (N) (mg/kg)	2008	0.3 (0)	0.3 (0)	0.4 (0)	0.538
	2009	0.4 (0)	0.4 (0)	0.4 (0)	0.965
Available ammonium (NH_4^+) (mg/kg)	2008	5.0 (2)	3.6 (1)	4.4 (4)	0.356
	2009	4.3 (3)	3.7 (2)	3.2 (1)	0.330
Available nitrate (NO_3^-) (mg/kg)	2008	6.2a (5)	3.6b (3)	9.7c (7)	0.011
	2009	6.1 (4)	5.3 (5)	7.3 (7)	0.643
Available phosphate (PO_4^-) (mg/kg)	2008	9.8 (3)	9.0 (2)	9.4 (2)	0.637
	2009	12.3 (3)	12.7 (4)	12.2 (4)	0.923
Available potassium (K) (mg/k^{10})	2008	250 (93)	252 (80)	223 (101)	0.586
	2009	308 (1111)	289 (89)	239 (83)	0.125
pH	2008	7.4 (1)	7.5 (1)	7.3 (1)	0.761
	2009	7.5 (1)	7.7 (1)	7.4 (1)	0.419
Electrical conductivity (EC) (ds/m^2)	2008	0.2 (0)	0.3 (0)	0.3 (0)	0.540
	2009	0.2 (0)	0.3 (0)	0.3 (0)	0.690

leaf length increased as NH_4^+ concentrations rose ($r^2 = 0.26$, $p < 0.001$) and increased as K decreased ($r^2 = 0.50$, $p < 0.001$).

F. hallii shoot biomass among sites was negatively correlated with prior- and current-year pH ($r^2 = 0.52$, $p < 0.001$; $r^2 = 0.53$, $p < 0.001$) and EC ($r^2 = 0.40$, $p < 0.001$; $r^2 = 0.44$, $p < 0.001$; Fig. 5). *P. pratensis* shoot biomass correlated positively to current-year pH ($r^2 = 0.13$, $p = 0.020$) and EC ($r^2 = 0.21$, $p = 0.002$; Fig. 5).

Table 5
Mean (±1 SD) soil properties of *Festuca hallii* treatments in 2008 and 2009 analyzed by ANOVA and Tukey post-hoc tests. Letters indicate significant differences at $p \leq 0.05$.

	Year	Site name			p
		Ellerslie	Byemoor	Drumheller	
June soil water (%)	2008	24.0a (6)	34.0a (5)	5.3b (1)	<0.001
	2009	8.7a (2)	6.5 (2)	5.3b (1)	<0.001
July soil water (%)	2008	15.0 (5)	17.0 (6)	16.0 (4)	0.326
	2009	31.0a (2)	27.0b (5)	13.0c (2)	<0.001
August soil water (%)	2008	16.0a (2)	11.0b (4)	5.2b (2)	<0.001
Total available soil water[a]	2008	46.0 (15)	60.0 (14)	41.0 (8)	0.001
	2009	40.0 (2)	34.0 (6)	19.0 (3)	<0.001
C:N ratio	2008	9.0a (1)	14.0b (5)	11.0a (1)	<0.001
	2009	11.0a (0)	10.0ab (0)	10.0b (1)	0.017
Total carbon (C) (mg/kg)	2008	4.2a (0)	2.6b (1)	2.6b (0)	<0.001
	2009	5.4a (0)	2.6b (0)	2.8b (0)	<0.001
Total nitrogen (N) (mg/kg)	2008	0.47a (0)	0.2b (0)	0.2c (0)	<0.001
	2009	0.51a (0)	0.3b (0)	0.3b (0)	<0.001
Available ammonium (NH_4^+) (mg/kg)	2008	5.6ab (0)	2.8a (1)	6.1b (5)	0.003
	2009	5.0a (1)	3.3b (2)	1.9c (0)	<0.001
Available nitrate (NO_3^-) (mg/kg)	2008	8.1 (2)	6.9 (15)	4.8 (6)	0.585
	2009	9.5 (5)	9.0 (15)	7.5 (4)	0.846
Available phosphate (PO_4^-) (mg/kg)	2008	8.4a (1)	8.4b (3)	12.4c (6)	0.004
	2009	12.0 (4)	12.0 (3)	15 (4)	0.077
Available potassium (K) (mg/k^{10})	2008	13.3a (1)	27.5b (4)	28.9b (9)	<0.001
	2009	15.2a (2)	26.3b (5)	29.3b (6)	<0.001
pH	2008	6.7a (0)	8.4b (0)	7.2c (0)	<0.001
	2009	6.7a (0)	8.4b (0)	7.5c (0)	<0.001
Electrical conductivity (EC) (ds/m^2)	2008	0.1a (0)	0.5b (0)	0.2a (0)	<0.001
	2009	0.1a (0)	0.4b (0)	0.2a (0)	<0.001

[a] Total available soil water in 2008 is June, July and August; 2009 is June and July.

Table 6
Mean (±1 SD) soil properties of *Poa pratensis* treatments in 2008 and 2009 analyzed by ANOVA and Tukey post-hoc tests. Letters indicate significant differences at $p \leq 0.05$.

	Year	Site name			p
		Ellerslie	Byemoor	Drumheller	
June soil water (%)	2008	21.0ab (4)	34.0b (4)	17.0a (5)	<0.001
	2009	6.9a (2)	5.8ab (1)	5.2b (2)	0.025
July soil water (%)	2008	16.0 (5)	18.0 (5)	14.0 (4)	0.129
	2009	34.0a (1)	26.0b (3)	13.0c (2)	<0.001
August soil water (%)	2008	18.0a (5)	9.7b (3)	4.8c (2)	<0.001
Total available soil water[a]	2008	46.0a (18)	60.0b (9)	36.0a (9)	<0.001
	2009	41.0a (2)	32.0b (3)	17.0c (5)	<0.001
C:N ratio	2008	11.0 (3)	12.0 (3)	10.0 (1)	0.256
	2009	11.0a (0)	10ab (1)	9.8b (1)	0.016
Total carbon (C) (mg/kg)	2008	4.9a (1)	2.7b (1)	3.3b (1)	<0.001
	2009	5.3a (0)	2.9b (0)	3.2b (0)	<0.001
Total nitrogen (N) (mg/kg)	2008	0.4a (0)	0.2b (0)	0.3c (0)	<0.001
	2009	0.5a (0)	0.3b (0)	0.3b (0)	<0.001
Available ammonium (NH_4^+) (mg/kg)	2008	4.7 (0)	4.2 (4)	4.5 (2)	0.857
	2009	5.2a (0)	3.7b (3)	2.3b (0)	0.001
Available nitrate (NO_3^-) (mg/kg)	2008	9.8a (5)	6.0ab (6)	4.7b (5)	0.038
	2009	3.0a (2)	7.5b (7)	8.3b (3)	0.019
Available phosphate (PO_4^-) (mg/kg)	2008	9.0 (1)	9.3 (2)	10.0 (3)	0.386
	2009	12.0 (3)	12.0 (3)	12.0a (4)	0.943
Available potassium (K) (mg/kg^{10})	2008	14.0a (2)	25.5b (6)	32.9c (7)	<0.001
	2009	15.8a (3)	30.3b (7)	35.9c (5)	<0.001
pH	2008	6.7a (0)	8.3b (0)	7.0a (0.4)	<0.001
	2009	7.0a (0)	8.3b (0)	7.0a (0.3)	<0.001
Electrical conductivity (EC) (ds/m^2)	2008	0.1a (0)	0.5b (0)	0.2a (0)	<0.001
	2009	0.1a (0)	0.4b (0)	0.1a (0)	<0.001

[a] Total available soil water in 2008 is June, July and August; 2009 is June and July.

5. Discussion

As hypothesized, *F. hallii* benefitted from straw amendment, possibly a result of added nutrients, such as K, known to leach from decaying straw (Christensen, 1985). The mulching property of straw may have assisted *F. hallii*, similar to Johnston's (1961b) observation that *Festuca campestris* seedlings produced the largest plants when grown on top a 5 cm litter layer. Straw treatments had little effect on *P. pratensis*, contrary to our hypothesis regarding its response to nitrogen immobilization. Reever-Morghan and Seastedt (1999) found similar results adding sucrose and sawdust to buried utility cable disturbances, noting little change in undesirable species. Our hypothesis about negative effects of disturbed soil on *F. hallii* recovery was confirmed. Byemoor had the highest pH and EC, possibly a result of disturbance during restoration (Hammermeister et al., 2003), and adversely affecting *F. hallii* growth. *P. pratensis* may not be affected by soil disturbance as it responded well to higher pH and EC. This could partially account for colonization of disturbed sites by *P. pratensis* (Brown, 1997; Slogan, 1997).

Our results of better *F. hallii* growth at the driest site, Drumheller, are similar to those of Stout et al. (1981) who recorded greater

Table 7
Mean (±SD) soil properties of undisturbed grassland at Byemoor and Drumheller sites.

	Site name	
	Byemoor	Drumheller
Total carbon (C) (mg/kg)	3.7 (1)	3.3 (1)
Total nitrogen (N) (mg/kg)	0.3 (0)	0.3 (0)
Available ammonium (NH_4^+) (mg/kg)	6.4 (3)	5.5 (2)
Available nitrate (NO_3^-) (mg/kg)	2.9 (1)	1.9 (0)
Available phosphate (PO_4^-) (mg/kg)	4.0 (2)	5.0 (1)
Available potassium (K) (mg/kg)	25.3 (8)	22.0 (4)
pH	6.4 (1)	6.6 (1)
Electrical conductivity (EC) (ds/m^2)	0.2 (0)	0.3 (0)

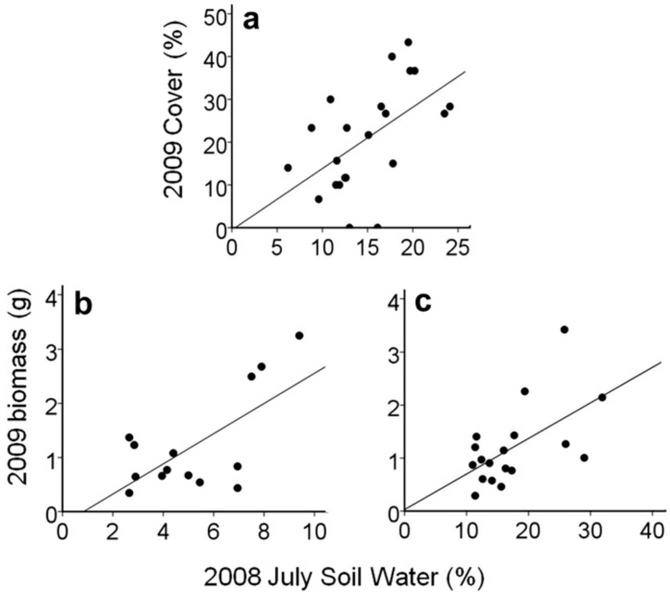

Fig. 2. *Festuca hallii* relationship to prior-year soil water, of **a** percent cover at Ellerslie ($r^2 = 0.30$, $p < 0.001$), and biomass at **b** Drumheller ($r^2 = 0.39$, $p < 0.001$) and **c** Byemoor ($r^2 = 0.33$, $p < 0.001$).

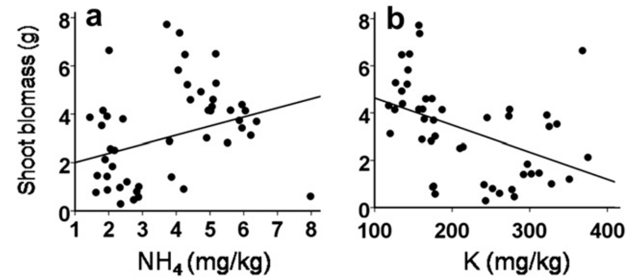

Fig. 4. Relation of *Festuca hallii* shoot biomass at all field sites to current-year soil **a** ammonium (NH_4) ($r^2 = 0.13$, $p = 0.026$) and **b** potassium (K) ($r^2 = 0.19$, $p = 0.002$).

F. campestris shoot biomass at the drier of two sites. We also discovered *F. hallii* positively responded to prior-year soil water. Mid-summer is typically when perennial grasses, in northern grasslands, produce and store carbohydrates for the next spring growth cycle (Menke and Trlica, 1981); therefore, abundant soil water would contribute to the following year's increases in shoot biomass, root biomass and leaf length. In our study, *P. pratensis* soil water responses were consistent with its growth habits, generally preferring mesic habitat and annual growth occurring in early summer (Sinton et al., 1996; Tannas, 2001).

Ammonium is immobilized quickly during straw decomposition (Smith and Douglas, 1971), which could account for decreases at all sites between 2008 and 2009. Smith et al. (1968) found a positive response of *F. campestris* to nitrogen fertilizer, which may explain the positive response of *F. hallii* to NH_4^+. All treatments were rototilled, which may have resulted in mineralization and nitrification (Dowdell and Cannell, 1975; Doran, 1980; Malhi et al., 1990). Andren and Paustian (1987) found that NO_3^- is released as straw decomposes; therefore, the combination of soil nitrification and straw decomposition probably accounted for increasing NO_3^- between 2008 and 2009. *P. pratensis* responded positively to higher nitrate, likely selecting NO_3^- rather than NH_4^+ (Darrow, 1939). Tilman and Wedin (1991) noted an inverse relationship between *P. pratensis* root biomass and soil NO_3^- concentrations. Our findings of longer leaf length with lower NO_3^- imply *P. pratensis* leaf growth incorporated NO_3^-.

Potassium readily leaches from straw as it decomposes, and similar amounts are found in wheat and barley straw (Christensen, 1985). *P. pratensis* treatments had significantly higher K than *F. hallii* treatments, indicating potential K uptake by *F. hallii*. Available K was significantly lower at Byemoor than the other two sites, possibly indicating soil disturbance (de Jong and Button, 1973; Hammermeister et al., 2003). No references were found regarding *F. hallii* use of K, although Templeton and Taylor (1966) found little difference in yield of *Festuca arundinacea* with varying concentrations of K fertilizer. Recommended K concentrations for *P. pratensis* is 160–240 ppm, whereas over 300 ppm is considered high (Robinson, 1985). Monroe et al. (1969) found *P. pratensis* clipping and root biomass, tiller density, leaf blade width and rhizome length were greater with up to 200 ppm of added K and less at 400 ppm. Byemoor and Ellerslie were within these concentrations, whereas Drumheller had over 300 ppm and significantly lower *P. pratensis* 2009 shoot biomass and leaf length.

Soil pH and EC increases during well site construction at Byemoor were likely due to soil horizon mixing (Hammermeister et al., 2003).

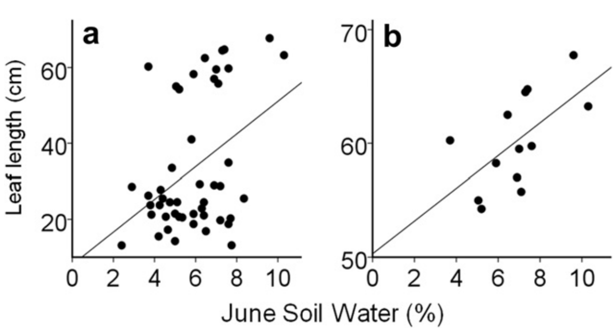

Fig. 3. Relation of *Poa pratensis* leaf length to current-year soil water in June, at **a** all sites ($r^2 = 17$, $p = 0.005$) and at **b** Ellerslie ($r^2 = 0.37$, $p < 0.001$).

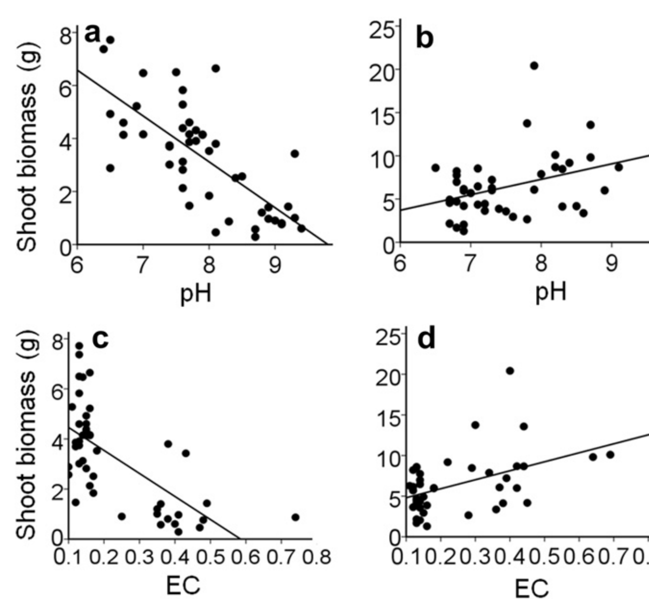

Fig. 5. Response of 2009 *Festuca hallii* biomass to soil **a** pH ($r^2 = 0.53$, $p = 0.020$) and **c** EC ($r^2 = 0.44$, $p < 0.001$) and *Poa pratensis* biomass and soil **b** pH ($r^2 = 0.13$, $p = 0.020$) and **d** EC ($r^2 = 0.21$, $p = 0.002$). Byemoor had pH values of 8 and above, and Drumheller and Ellerslie had pH values below 7.5.

No studies were found reporting changes in rough fescue growth in soil pH over 8. Johnston et al. (1971) found *F. campestris* cover was reduced if soil pH increased from 5.7 to 6. With higher pH, *P. pratensis* had greater leaf length and number, root and leaf weights and number of rhizomes (Darrow, 1939); increased vigor, density and color (Skogley and Ledeboer, 1968); and greatest abundance (Tilman and Olff, 1991). Nevertheless, these studies analyzed *P. pratensis* reaction to acidic soils, and did not assess reaction to pH higher than 7, which we found at Byemoor.

6. Conclusion

Although our results for *F. hallii* were best for the high-straw amendment, low straw also had better results than no amendment, leading us to conclude addition of straw as a soil amendment is a possible solution to poor establishment of *F. hallii*. Field experiments are subjected to numerous uncontrolled effects on soil properties and vegetation growth. Temperature, available water, soil microbial actions, vegetation competition, to name a few, could account for variations in vegetation growth. Nevertheless, this study illustrated the importance of reducing disturbances in native grassland, avoiding soil disturbance or mineralization that might attract non-native species and assist in their establishment. Straw is inexpensive and readily available and was proven to assist a native bunch grass. Incorporated straw increases soil nutrients, such as NH_4^+ and K, and provides surface mulching that may assist other native species establishment. If soil must be disturbed, the threat to remnant grasslands may be reduced if soil amendments targeted at native species are incorporated.

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